

**AUTO REPAIR HUB: A WEB-BASED BOOKING AND MANAGEMENT
PLATFORM**

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CHAPTER 1

INTRODUCTION

Project Context

In an era where digital solutions are reshaping industries, the auto repair sector in many rural areas including Virac, Catanduanes remains bound to manual and outdated methods. Shops such as Molje Lube, Precision Tech Motoshop, and CT Gear continue to operate using walk-in service requests, handwritten appointment logs, and in-person or phone-based communications. This traditional system presents major challenges in terms of efficiency, scheduling, and customer satisfaction. A study by Bughin et al.,2021, digital technologies are reshaping customer expectations and operational efficiency across various sectors, including automotive services. Similarly, Deloitte (2021), highlights that digital platforms significantly enhance customer engagement of auto repair shops, especially in provincial areas such as Virac, Catanduanes, that continue to operate using traditional, manual records.

Currently, customers must either call or physically visit a shop to check for availability or inquire about the progress of ongoing services. Job orders are manually logged, making it prone to overlaps, forgotten requests, and miscommunication. Furthermore, with no clear system for queue management or service tracking, clients are left uncertain about their wait time or when their vehicle will be ready. The lack of digital

visibility makes it difficult for both customers and shop staff to stay informed and organized.

To resolve these operational bottlenecks, the proposed Auto Repair Hub introduces a centralized, web-based platform that digitizes the booking and service tracking process while enhancing customer communication. Key features of the system include, GIS Location Map, where customers can view a live map displaying nearby shops such as Molje Lube, Precision Tech Motoshop, and CT Gear, allowing them to select the most accessible shop based on their location. Online Booking System, where customers can schedule repair or maintenance appointments at their convenience without the need to visit the shop in person to inquire. Service Tracker Queue where customers will be able to see their position in the service queue and receive updates on their vehicle's status, reducing uncertainty and frustration. And lastly, SMS Notification, where the system will send automated text messages to inform clients about appointment confirmations, service updates, and completion notices keeping them engaged and informed throughout the process. By implementing these integrated features the aAuto Repair Hub eliminates the inefficiencies of traditional manual processes. Customers no longer need to rely on uncertain phone calls or time-consuming visits, as the online booking system offers availability and scheduling. The GIS Location Map provides instant visibility of nearby service centers, addressing the issue of accessibility. Meanwhile, the service tracker queue and SMS notifications directly mitigate customer frustration by offering transparent, timely updates, improving trust and satisfaction. For

shop staff, the system reduces manual workloads, minimizes scheduling conflicts, and ensures that all job orders are accurately logged and tracked. Collectively, these features establish a seamless, technology-driven experience that addresses the root causes of delays, miscommunication, and low customer engagement in local repair services.

This platform directly addresses the service inefficiencies that plague local auto shops. It enhances customer satisfaction by offering more control and visibility, while shop personnel benefit from a more organized and streamlined process. The system will be developed for auto repair shops within Virac, Catanduanes, particularly, Molje Lube, Precision Tech Motoshop and CT Gear. The main users of the platform include vehicle owners (customers) who will book services online, receive updates via SMS, view shop locations through GIS, and track their queue status. Shop staff who will manage service entries, update job order status, and respond to online bookings. Shop owners who will supervise daily operations, track performance, and ensure better customer service delivery using a digital dashboard.

The development of Auto Repair Hub is a timely and strategic solution for modernizing the workflow of auto repair shops in Virac. By integrating GIS-based location services, service tracking queue, and automated SMS notifications, the system empowers both customers and service providers. This initiative not only improves local service quality but also advances the digital transformation of small-to-medium enterprises in the automotive sector of Catanduanes.

Purpose and Description

The primary purpose of the Auto Repair Hub project is to design and develop a centralized, web-based platform that enhances the efficiency, accessibility, and transparency of auto repair services in Virac, Catanduanes. Specifically, the system addresses the challenges faced by local auto repair shops Molje Lube, Precision Tech Motoshop, and CT Gear which currently rely on traditional, manual methods for appointment scheduling, customer communication, and service tracking.

The platform will enable customers to book appointments online, locate the nearest partner shop through an integrated GIS location map, and monitor the status of their vehicle repairs using a Service Tracker Queue. Additionally, the system will feature SMS notifications that automatically inform customers about booking confirmations, repair progress, and job completion. For shop owners and staff, the platform will simplify service management by providing a centralized dashboard for organizing job orders, tracking active services, and avoiding booking conflicts. It also helps, reduce manual errors, and improve communication between staff and clients.

The project aims to bridge the gap between traditional auto repair operations and modern customer service expectations by delivering a digital solution that is scalable, user-friendly, and tailored to the local context of Virac. Ultimately, the system is expected to contribute to better time management, increased customer satisfaction, and higher operational productivity for auto repair businesses in the region.

Objectives

To develop Auto Repair Hub, a web-based booking and management platform for selected auto repair shops in Virac, Catanduanes, that streamlines appointments, enables service tracking, GIS-based location, and SMS notifications

Specifically, the objectives are the following:

1. To build a functional and reliable online booking system with SMS alerts, ensuring accurate operations and consistent performance.
2. To integrate a GIS location mapping for finding nearby partner shops, enhancing user accessibility and experience.
3. To implement a Service Tracker Queue, promoting service transparency and timely updates.
4. To evaluate the developed system via end users using the following ISO/IEC 25010 parameters:
 - a) Functional Suitability
 - b) Performance Efficiency
 - c) Compatibility
 - d) Interaction Capability
 - e) Reliability
 - f) Security
 - g) Maintainability
 - h) Flexibility

i) Safety

Scope and Limitation

The system aims to digitize and streamline core service functions, including appointment scheduling, service tracking, and customer notifications. Its key features include an online booking system, a service tracker queue to monitor job status, a GIS location map for finding nearby shops, and an SMS notification system for sending automated updates regarding bookings and service progress. Additionally, an administrative dashboard will allow shop owners and staff to manage appointments, monitor ongoing services, and facilitate customer engagement. The system will be accessible via desktop and mobile web browsers and will require users to log in for access to its features. The Auto Repair Hub is a web-based platform designed to serve selected auto repair shops in Virac, Catanduanes, specifically Molje Lube, Precision Tech Motoshop, and CT Gear.

However, the system has several limitations. It does not include inventory or spare parts management, as the project is focused solely on service scheduling and customer communication. The platform is currently limited to the three identified shops and is not designed to support broader business integration at this stage. The full functionality of the system, particularly the GIS features and SMS alerts, requires a stable internet connection. Furthermore, the platform does not support online payment transactions, and

all customer and shop accounts may require manual approval by the system administrator to ensure security and data accuracy.

Definition of Terms

The following terms are defined based on they will be used in this project.

Admin Dashboard is the interface used by shop administrators to manage bookings, monitor active services, view queue statuses, and communicate with customers.

Auto Repair Hub is a web-based booking and service management platform created for selected auto repair shops in Virac, Catanduanes, offering digital appointment scheduling, service queue tracking, and customer notifications.

Auto Repair Shop refers to local vehicle service providers in Virac, Catanduanes, which we identified as still using manual processes for booking and service tracking.

Booking System is a digital interface that allows customers to schedule appointments online.

Compatibility is the ability of the system to operate and exchange data with other systems or devices, particularly across different web browsers and screen sizes.

Functional Suitability is the degree to which a system or software product provides functions that meet stated and implied needs when used under specified conditions.

Flexibility is the degree to which a system can be adapted efficiently and effectively to changes in its requirements, environment, or context of use.

Geographic information System (GIS) is a computer-based system designed to capture, store, manipulate, analyze, manage, and present all types of geographically referenced data.

Interaction Capability refers to the degree to which a system allows users to interact with it effectively, efficiently, and with satisfaction, using available input/output devices and modalities.

ISO/IEC 25010 Provide a structured framework to define, measure, and evaluate the quality of software products and systems.

Maintainability is about how easily a system can be updated, improved, or changed to fix problems or add new features. This is made possible by using modular coding, clear documentation, and organized database design. These practices help developers and system administrators quickly adapt the platform to future requirements or add new functions, like payment systems or wider shop coverage.

Location Mapping is the integration of geographic information to display shop locations on a map.

Performance Efficiency is the system's ability to respond quickly and operate effectively in terms of resource use, ensuring real-time updates in the service queue and fast loading of the GIS map.

Reliability is the capability of the system to maintain performance under normal and unexpected conditions without failure or loss of data, ensuring stable service for users.

Safety is the degree to which a system operates in such a way that mitigates the risk of harm to people, property, or the environment.

Security is the protection of system data and operations against unauthorized access, including secure logins, account approvals, and data privacy.

Service Tracking Queue is monitoring the progress and status of the vehicle service from booking to completion.

Short Message Service (SMS) notification refers to alerts that appear within a mobile app or web application while the user is actively using it.

Web-Based Booking and Management Platform is an online system that allows users to schedule services and enables administrators to manage bookings, monitor schedules, and track operations through a centralized, accessible interface.

Conceptual Framework

This section shows the flow and interaction of system components that guide the development of the Auto Repair Hub: A Web-Based Booking and Management Platform. It shows how user inputs such as service bookings and customer data are processed through features like service queue tracking, GIS-based shop location, and SMS notifications. This framework ensures that the system meets its goal of improving service efficiency and customer experience for auto repair shops in Virac, including Molje Lube, Precision Tech Motoshop, and CT Gear.

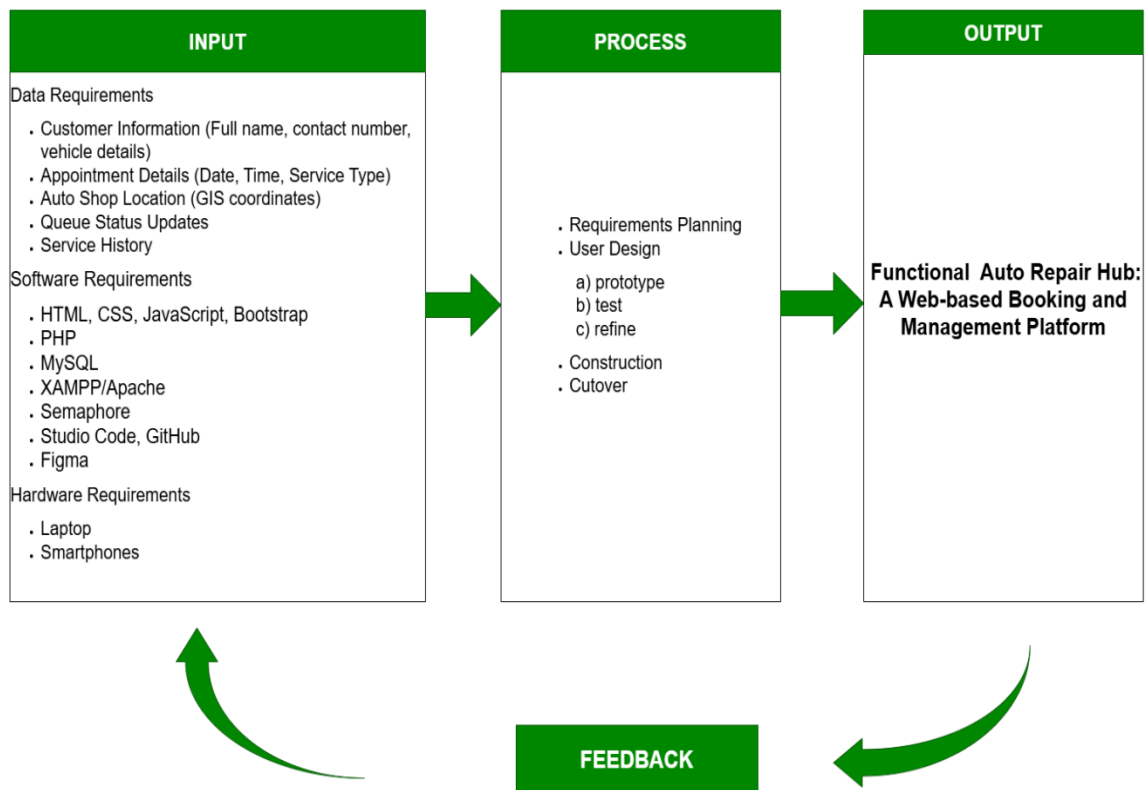


Figure 1. Conceptual Framework

This Figure 1 represents the Conceptual Framework The Input section describes the basic components required to build the system. It is divided into three categories: Data Requirements, Software Requirements, and Hardware Requirements. The Data Requirements detail the key information the system must collect and manage, including customer details (such as full name, contact number, and vehicle information), appointment details (like date, time, and type of service), the auto shop's location using GIS coordinates, queue status updates, and service history. The Software Requirements list the technologies and tools used for development, including HTML, CSS, JavaScript, and Bootstrap for the frontend; PHP and MySQL for the backend; and tools such as XAMPP/Apache for local hosting, Semaphore for CI/CD automation, Visual Studio Code and GitHub for coding and version control, and Figma for UI/UX design. The Hardware Requirements cover laptops for development and smartphones for testing or accessing the system on mobile devices.

The Process section outlines the steps involved in developing the system, following a clear methodology. These steps include gathering requirements, designing the system architecture and user interface, coding and development, creating conceptual models (like data flow and entity-relationship diagrams), testing the system to find bugs or issues, deploying the system for user access, and reviewing its performance for improvements.

The Output represents the expected result of a fully functional Auto Repair Hub that enables users to book services online, find partner shops through GIS, and monitor their service queue status. This digital solution is designed to modernize and simplify the daily operations of auto repair shops in Virac, Catanduanes. Finally, the Feedback loop at the bottom of the conceptual framework diagram shows the system's ongoing improvement process. It emphasizes that the system will evolve by incorporating user feedback and performance evaluations to ensure continuous enhancement and responsiveness to user needs.

CHAPTER 2

REVIEW OF RELATED LITERATURE/SYSTEMS

This chapter presents related studies that support the development of the Auto Repair Hub: A Web-Based Booking and Management Platform. It reviews existing literature on web-based booking systems, GIS integration, service queue tracking, and SMS notifications. The review identifies gaps in the current manual processes of local auto repair shops in Virac such as Molje Lube, Precision Tech Motoshop, and CT Gear and establishes the relevance of the proposed platform in improving efficiency and customer service.

Auto Repair Vehicle Management Systems

Auto repair vehicle management systems streamline the administration of automotive service operations by automating records of vehicle details, service histories, and customer profiles. These systems reduce manual workloads, minimize errors, and enhance overall shop productivity. Their adoption is essential for modern workshops aiming to improve service efficiency and customer satisfaction.

Car owners often forget their service schedules and have trouble keeping accurate records, which can lead to poor vehicle maintenance, safety risks, and higher costs over time. Traditional reminders, like stickers on the windshield, can fade or fall off, making

them unreliable. Many digital solutions are limited to certain car brands or require downloading an app, which can be inconvenient. To solve these problems, Raman et al. (2021) developed the Automobile Service E-Assistant, a browser-based system where users can store service records digitally, get automated maintenance reminders by email (such as for oil changes or tire replacements), find nearby workshops, and view their full service history all without needing to install an app. The system was built using the Waterfall model with PHP, MySQL, HTML/CSS, and JavaScript. Its usability was tested with 30 car owners through the WAMMI questionnaire, which assessed attractiveness, controllability, efficiency, helpfulness, and learnability. The results were positive: 93% of users found the system helpful for keeping track of service schedules, and 80–83% gave good ratings for efficiency and learnability. Some issues were noted, such as confusing password recovery that opened in new tabs and a lack of introductory guidance for new users. Interestingly, 73% of participants had never used similar apps before, and 60% preferred local workshops over branded dealerships. The system's strengths include its web-based design that works on any device, high user satisfaction, and thorough usability testing with the WAMMI framework. However, it has some drawbacks, like overlapping features with existing tools such as MyMazda, missing mobile optimization, and weak data security measures, which raise concerns about its ability to scale and protect user data effectively.

Gomathy, Chandrasekhar, Mallikarjun, and Geetha (2022) evaluated the daily operations of auto repair shops and found that many shops still rely on manual processes for managing customer records, repair tracking, and billing. This manual system contributes to inefficiencies and delays in responding to client inquiries. Their research highlights the need for automated vehicle service management systems to streamline operations and improve customer service. These findings underscore the relevance of digitizing shop workflows, a principle that Auto Repair Hub also adopts through its integrated, user-friendly platform for managing appointments, repairs, and client communications.

Damiato and Utami (2021) proposed an Android-based online car repair shop application focused on home service functionality. With millions of private car users in Indonesia, the app addresses the need for mobile service delivery in cases where users face emergencies like tire punctures or accidents in remote locations. The application allows users to locate and request a nearby mobile mechanic without needing to seek help manually, providing convenience and safety. This research supports the value of mobile repair service integration, echoing the goals of Auto Repair Hub in offering diagnostics and emergency.

The automotive repair and maintenance industry has experienced a paradigm shift from manual processes to automated, technology-driven systems aimed at improving operational efficiency and customer satisfaction. Gomathy et al. (2022) highlighted that many auto repair shops still rely on manual processes for managing customer records,

repair tracking, and billing, which leads to slow service delivery and poor client engagement. This growing inefficiency has pushed developers and researchers to propose digitized solutions. Ji (2021) designed a structured management system based on the Struts2 framework that incorporates modular functionality, user-friendly interfaces, and layered architecture to enhance process automation and enterprise oversight in repair services. Similarly, Parmar, Bansal, and Sahatiya (2024) advocated for the development of mobile applications tailored to the repair and maintenance needs of vehicle owners, leveraging the growing accessibility of smartphones to bring services closer to users and facilitate seamless maintenance workflows. Together, these studies emphasize the pressing need for integrated, intelligent platforms that support both operational management and customer interaction in the auto service industry.

The study by Chandra et al. (2023) addresses the challenges travelers face during vehicle breakdowns, such as difficulty locating nearby garages, navigating unfamiliar areas, and finding reliable mechanics, by developing an Android application that utilizes GPS and Google Maps API to connect users with nearby service providers. The proposed solution enables users to request repair services, allows garages to accept jobs and issue digital bills, and incorporates features like advance booking, emergency contacts, and an online marketplace for automotive parts. Built using Android Studio and Firebase for real-time database management, the system is structured into user, garage, and admin modules, supported by GPS tracking and a 24/7 helpdesk. The authors highlight the app's effectiveness in streamlining service requests and improving user convenience, though

they acknowledge the need for further testing and scalability analysis. Future enhancements, such as vehicle health monitoring and integration with towing services, are suggested to expand the app's utility, positioning it as a promising tool for modern automobile servicing.

One relevant study is the project titled Vehicle Service Center Management System developed by Monisha (2022), which aims to automate the operations of a traditional vehicle service center through a web-based application. The study highlighted the need to move from manual, paper-based processes to a digital system that supports online booking, service tracking, and customer communication. The system was designed to improve efficiency, accuracy, and customer engagement by allowing users to book services remotely, while the business could manage records such as income, expenses, and service history on a centralized platform.

Monisha's system used ASP.NET for front-end development and C# for back-end logic, running on a Windows-based environment. The project showed how an online platform can simplify managing vehicle services by automating scheduling, order tracking, and database maintenance. It also included common software engineering practices like unit testing, integration testing, and prototyping. Importantly, the study identified several benefits, such as better customer interaction, fewer operational delays, and improved data accuracy. This capstone project follows similar principles in system design, especially the use of web-based technologies and database integration. However,

it builds on Monisha's work by adding features like mobile responsiveness, SMS notifications, and a geographic information system (GIS) to map repair shops, tailored to the needs of automotive businesses in Virac, Catanduanes.

Several studies have highlighted the shift from manual to automated auto repair processes to improve efficiency, service tracking, and customer satisfaction. Raman et al. (2021) developed a browser-based system that provides email reminders, digital service records, and workshop locators, helping car owners avoid forgetfulness and poor recordkeeping. Likewise, Monisha (2022) showed the benefits of managing service centers online through web platforms with appointment scheduling and customer communication features. Gomathy et al. (2022) pointed out that manual documentation causes inefficiencies, emphasizing the need for automation in repair tracking and billing. Many existing systems lack important features such as mobile responsiveness, real-time notifications, and GIS integration. These features are especially important in rural areas like Virac, Catanduanes, where internet access can be unreliable and finding physical shops is often difficult. To address this, the Auto Repair Hub offers a mobile-friendly web interface, a GIS-based shop locator for easier access, and SMS alerts to provide real-time updates without depending entirely on the internet. Together, these features overcome the shortcomings found in previous systems and ensure that the platform is both accessible and practical for rural communities, making it a more inclusive and relevant solution.

Booking and Queuing Systems

Efficient booking and queuing mechanisms play a vital role in enhancing customer experience and optimizing the operational flow in automobile repair shops. Several studies have emphasized the importance of integrating intelligent scheduling and queue management systems to reduce wait times, eliminate idle labor, and ensure effective service delivery.

In Virac, Catanduanes, most car service centers still use manual methods for booking appointments, leading to inefficiencies and longer wait times. Dalisay (2019) addressed this issue in the Philippine context by developing a multi-platform online reservation system, allowing customers to book services via mobile or web. Integrating similar technology into the Auto Repair Hub project can streamline operations, reduce scheduling conflicts, and improve service delivery for both customers and mechanics.

Katsman et al. (2023) applied queuing theory to model multi-stage, multi-component repair systems. Their simulation used stochastic modeling to identify optimal configurations for service stations. The findings revealed that integrating mathematical modeling into workshop logistics can reduce bottlenecks and increase throughput by forecasting service durations and queue densities. Similarly, Wong (2020) introduced an online booking system for a local mechanic shop that previously relied on walk-in services and phone calls. His mobile-based system enabled customers to book services in advance and receive automated reminders, leading to better time management, improved customer retention, and reduced manual workload.

Existing user management systems for remote laboratories often struggle with compatibility and scheduling, which leads to long wait times and underused resources (Khazri et al., 2019). To solve this, Khazri et al. (2019) created the ELAB FSBM platform, which combines a dual reservation system for calendar bookings with a priority queue for on-demand access. They used UML modeling to design role-based workflows for visitors, users, and administrators, all implemented in a web-based system that hosts experiment documentation and simulations. This solution provides structured access control, allowing users to reserve time slots or join queues, while dynamically assigning labs based on availability and guaranteed session lengths. Although the platform shows promise in managing both planned and spontaneous access, do not offer data on performance improvements or scalability, so further testing is needed.

The study by Guevarra and Ignacio (2021) tackles the problem of outdated and inefficient online search methods for finding nearby auto repair shops, especially in emergencies. Users often struggle with inaccurate or missing service information, outdated contact details, and slow website navigation, which can cause stress and waste time. To address this, the authors developed an Android app called "OTO Direct" that uses location-based services to help users quickly find and connect with nearby auto repair shops. The app includes a user-friendly interface, maps, categorized services, and direct contact information for shops. The development followed a prototyping approach, involving stages such as requirements analysis, design, initial testing, user feedback, and final implementation. It was tested with 35 participants, including auto shop owners and

car enthusiasts, and evaluated on user interface, system performance, integrity, and security. The results showed high user acceptance, with average ratings from "Very Good" to "Excellent" and an overall score of 4.56 out of 5. These findings suggest the app is an effective way to improve access to auto services using real-time, location-based data. This study offers a practical and well-designed prototype that meets a common, often overlooked need. Future enhancements like shop verification, online booking, and payment integration could further improve its usefulness and reach.

Based on the literature reviewed, queuing systems play a crucial role in managing customer flow and reducing wait times in service industries. At the Auto Repair Hub, this is achieved through a real-time Service Tracker Queue, which allows customers to see their service status (such as Pending, In Progress, or Completed) and their place in line. Staff update these statuses, making the process transparent and helping to ease customer frustration. The system also sends SMS notifications to keep customers informed even when they are not online. This setup shows how queuing principles can be applied effectively to create a smoother, more organized experience for both customers and staff.

Technologies integrated into auto repair systems

In this project, the integration of modern technology into auto repair services is a key focus of Auto Repair Hub. By incorporating advanced digital tools and systems, the platform aims to revolutionize traditional methods of booking, queuing and repairing

vehicles. This innovation enhances the accuracy and efficiency of repair operations while ensuring greater vehicle safety and reliability, meeting the growing demands of both auto repair professionals and their clients.

Bramley and Baker (2023) emphasize how modern technologies are transforming automotive repair shops by streamlining operations and improving service delivery. With the rising cost of living affecting both consumers and repair businesses, there is a pressing need for digital solutions that enhance efficiency, reduce overhead costs, and increase customer satisfaction. Their study highlights the importance of adopting operational technologies, new service models, and supportive software tools to maintain profitability and competitiveness. These insights are relevant to the development of Auto Repair Hub, which aims to provide a centralized platform that supports efficient booking, communication, and service tracking for both customers and auto repair shops. The integration of digital tools, as demonstrated in their findings, is essential for sustaining business operations and ensuring high-quality service in today's automotive service industry.

The automotive repair industry faces ongoing challenges such as unclear pricing, long service times, and a loss of trust between customers and repair shops. These issues are worsened by technicians working with outdated equipment and unfair pay systems. Liang suggests using three technologies together: (1) blockchain to ensure transparent pricing and parts tracking, (2) IoT sensors for real-time vehicle diagnostics, and (3) augmented reality (AR) to help customers better understand repairs. The study also

recommends forming partnerships between traditional garages and mobile repair startups. Using a mix of research methods observing shops, holding customer focus groups, and testing an AR diagnostic app the study looks at customer needs and pain points through established frameworks. Results show that AR greatly improved customers' understanding of repairs, but blockchain pricing models were met with skepticism without market comparisons. Mobile repair services were convenient but raised concerns about reliability. Liang's tech-driven approach addresses key industry problems, but challenges remain due to established practices and consumer hesitation with new technology. The study would benefit from including cost-benefit analyses and more pilot tests with a wider range of users.

The study by Aponsu et al. (2022) highlights important problems in the automobile service industry caused by manual paper-based systems. These issues include lost records, long wait times, and a lack of transparency, which ultimately reduce customer trust. To solve these problems, the researchers designed a web-based Automated Car Service Management System that offers online booking, digital payments, inventory tracking, and centralized service records. They used a mix of methods, including interviews with service managers and vehicle owners, along with technical development using C#, .NET, and MySQL. The result is a dual-dashboard system for both administrators and customers. The study shows that the system helps reduce wait times through online scheduling, improves efficiency with real-time inventory management, and increases transparency by providing access to service

histories. However, some challenges remain in moving service centers away from manual processes. Although the solution addresses key industry issues with its flexible design, the study could be improved by considering scalability, cybersecurity, and testing in regions beyond Sri Lanka. These steps would help make the system more widely applicable and open up opportunities for future improvements like AI features and better user experience.

The use of digital technologies has become crucial in modernizing auto repair services, especially in boosting efficiency, transparency, and communication with customers. Research highlights the value of online systems that handle booking automation, service tracking, and real-time updates. However, many current platforms lack features that are particularly helpful in rural areas, such as location mapping and communication that works offline. This study presents the Auto Repair Hub, which fills these gaps by using two main technologies: Geographic Information System (GIS) and Short Message Service (SMS). The GIS feature lets customers find and choose nearby partner shops through an interactive map, making it easier to access and decide, especially for those not familiar with the area. The SMS system sends real-time updates about service bookings and status changes, even in places with poor or unstable internet. By focusing on these practical and easy-to-use technologies, the Auto Repair Hub offers a cost-effective, customer-friendly solution that meets the needs of both customers and shop owners in rural communities like Virac, Catanduanes. These tools help improve transparency, efficiency, and customer satisfaction in local auto repair services.

ISO/IEC 25010 in Evaluating the Systems

The ISO/IEC 25010 standard outlines comprehensive criteria for evaluating software quality, including key attributes such as functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability (ISO/IEC, 2023). Specifically, focusing on usability, reliability, and functional suitability enhances the platform's ability to deliver efficient booking and service management for auto repair shops in Virac, Catanduanes. This quality-driven approach supports a user-centered system that meets the needs of both shop owners and clients.

El Fhel, Sardi, and Idri (2022) evaluated mobile GIS systems using the ISO/IEC 25010 quality model and emphasized that functional suitability and usability are significantly shaped by system design, especially features like online data access. Their study supports the use of international standards such as ISO/IEC 25023 to ensure system quality and stakeholder satisfaction. Similarly, Cai, Sun, and Dobbie (2022) applied ISO/IEC 25010 to assess formal software models, focusing on repairing unreachable states while preserving overall system quality. These findings highlight the importance of aligning the Auto Repair Hub with ISO/IEC 25010 to ensure it meets global standards in functionality, reliability, and usability particularly critical in delivering efficient, high-quality service in Virac, Catanduanes.

Quality assessment standards like ISO/IEC 25010 offer a framework for evaluating a system's performance in areas such as functionality, usability, security, and

reliability. Research by El Fhel et al. (2022) and Cai et al. (2022) supports the effectiveness of these criteria in ensuring software reliability and user satisfaction. However, only a few auto repair systems use formal ISO-based evaluations, and those that do rarely adapt the standards to fit local needs. There is little evidence of ISO-aligned assessments being carried out for rural automotive platforms or web-based booking systems in developing areas.

Synthesis

The reviewed literature shows a clear shift from manual to digital systems in auto repair services. Web-based platforms, queue management tools, and mobile booking apps have consistently improved operational efficiency, customer satisfaction, and service transparency. Systems like the Automobile Service E-Assistant and Vehicle Service Center Management System offer benefits such as centralized booking, digital records, and automated customer notifications. However, many existing systems still lack important features like GIS-based shop locators, real-time SMS updates, and offline access features that are especially important in rural areas like Virac, Catanduanes. Additionally, many studies assume users have stable internet access and are comfortable with technology, which limits their usefulness in underserved communities.

The Auto Repair Hub fills these gaps by integrating a GIS-powered shop locator, SMS communication, and a mobile-friendly design, all within a web platform accessible to both customers and shop staff. A unique feature of this system is its Service Tracker

Queue, which provides real-time updates on job progress. This feature is based on simple queueing principles, organizing service flow in a logical, first-come-first-served way. Although the system does not use formal mathematical queueing models, it applies queueing concepts to manage client flow, improve transparency, and reduce customer uncertainty during the service process.

The platform is also evaluated using ISO/IEC 25010 software quality standards, ensuring it meets criteria for functionality, usability, maintainability, and reliability. With its context-aware features and standards-based evaluation, the Auto Repair Hub stands out as a practical and scalable solution designed to support the digital transformation of auto repair shops in rural areas.

CHAPTER 3

TECHNICAL BACKGROUND

This chapter outlines the technical requirements necessary for the successful implementation of the Auto Repair Hub system. It includes hardware and software specifications, data requirements, and a breakdown of user requirements. The goal is to ensure the development and deployment of a reliable, scalable, and user-friendly system tailored for the auto repair hub in Virac, Catanduanes.

Hardware Requirements

The following table summarizes the essential hardware components required to develop, deploy, and run the Auto Repair Hub: A Web-Based Booking and Management System effectively.

Table 1. Hardware Requirements

Hardware Requirement	Specification	Description
Laptop	Intel(R) N100 800 MHz 8GB RAM, 512GB SSD	Serves as a central workstation for building the system from start to finish.
Smartphones	Android/iOS	Used by customers to access the system, make bookings, and receive SMS updates on service status.

Table 1 Hardware Requirements The Auto Repair Hub system needs some key hardware to work smoothly during development, deployment, and daily use. A laptop with an Intel N100 processor, 8GB of RAM, and a 512GB SSD serves as the main workstation for building and managing the system. Customers use smartphones with at least a quad-core processor, 3GB of RAM, and 4G or 5G LTE to easily access the platform, make bookings, and get real-time updates. These devices make the system accessible and user-friendly for everyone. Additionally, a 64GB USB 3.0 flash drive is used for backing up files, transferring data, and installing software when needed. Together, these hardware components create a cost-effective and efficient setup for the Auto Repair Hub, designed to support local auto repair shops and their customers.

Software Requirements

The table below presents the software tools and platforms used in the development and operation of the Auto Repair Hub system:

Table 2. Software Requirements

Software Requirements	Specification	Description
Operating System	Windows 11	Runs admin devices, ensuring system compatibility and stability.
Web Server	XAMPP / Apache	Hosts the system locally during development and testing.

Software Requirements	Specification	Description
Backend Language	PHP	Handles system logic and process server-side functions.
Database System	MySQL	
Frontend Technologies	HTML, CSS, JavaScript (with Bootstrap)	Builds the user interface to ensure responsiveness and usability.
SMS Notification System	Semaphore	Sends automated SMS updates to customers and admins.
Development Tools	Studio Code, GitHub	Used for coding (VS Code) and managing versions and collaboration (GITHUB).
Prototyping	Figma	Designs UI/UX layouts before coding to guide development.

Table 2 shows The Auto Repair Hub system relies on a range of software tools to support its development and operation. Windows 11 is used on admin devices for a stable and compatible environment. For development, XAMPP with Apache serves as the local web server, allowing the system to be hosted and tested efficiently. The backend logic is handled by PHP, while MySQL manages the system's database, storing essential data like user profiles, bookings, and service records. The frontend is built using HTML, CSS, and JavaScript with Bootstrap, creating a responsive and user-friendly interface. Semaphore is integrated to automate SMS notifications, keeping customers updated in real time. Development is carried out using Visual Studio Code for writing and editing code, while

GitHub supports version control and collaboration. Lastly, Figma is used to prototype and design the user interface, ensuring a clear and effective layout before actual implementation. Together, these tools contribute to a reliable and user-centered system.

Data Requirements

The following table defines the types of data needed for system development and functionality, along with how they are collected and processed:

Table 3. Data Requirements

Construct/ Variables	Source of Data	Data Gathering Technique	Instrument	Data Processing
Booking Information	Customers	Online Form	Registration Form	Stored in customer database
Customers Information	Customers	Online form, interviews	Questionnaire	Used to populate system dropdowns
Shop Information	Admin	Manual Entry	Admin Input Form	Displayed in system dashboard and locator
GIS Location	Google Maps API	API integration, Manual Entry	GPS Coordinates, Map View	Mapped and displayed through integrated shop locator
SMS Notifications Logs	System	Automatic Logging	SMS Gateway	logged and saved in the notification history
System Evaluation	End-Users	Questionnaires	Survey Questionnaire	1. Mean 2. Frequently count 3. Percentage

Table 3 shows the Data Requirements for the Auto Repair Hub, a web-based booking and management platform, ensuring that all functions run smoothly and provide accurate, user-focused services. Booking details are collected directly from customers through an online form, capturing important information like the chosen service, shop, date, and time. This information is saved in the customer database and is vital for managing appointments and service queues. Customer data, including personal and vehicle details, is gathered through online forms and interviews using structured questionnaires. This information helps fill system dropdowns and improves the overall user experience. Shop details, such as names and services offered, are entered manually by the admin using an input form and shown on the dashboard and shop locator for customers to see. GIS location data comes from Google Maps API integration and manual GPS input, enabling the system to display nearby partner shops on an interactive map. SMS notification logs are automatically recorded by the system through the integrated SMS gateway, tracking all service updates sent to customers for reference and accountability. Finally, system evaluation data is collected from users through Likert-scale surveys, with responses analyzed using simple statistics like averages and frequency counts. This helps measure user satisfaction and system performance, guiding future improvements. Together, these data inputs create a responsive, accurate, and data-driven system designed to meet the needs of auto repair businesses and their customers.

User Requirements

This section outlines the core modules and functionalities of the system, as well as the respective user roles that interact with each module:

Table 4. User Requirements

Module	Function	Description	User Role
Dashboard	Overview of System Status	Displays basic system summaries like information of customers, total jobs and pending bookings	Admin, Staff
Booking Module	Schedule Repair Services	Allows customers to see available time slots and send booking requests.	Customer
Service Management	Update Work Progress	Enables staff to update status and progress of auto repair hub services.	Admin, Staff
User Management	Account and Role Handling	Admin can manage staff accounts and assign permissions.	Admin
SMS Notification	Status Updates	Sends automatic SMS notifications for each job stage.	System (Automated)

Table 4 User Requirements shows The Auto Repair Hub: A Web-Based Booking and Management Platform relies on several types of data to work smoothly and provide accurate, user-friendly services. Customers enter booking details like the service they want, the shop, date, and time through an online form. This information is saved in the customer database and helps manage appointments and service queues. Customer details, including personal information and vehicle data, are collected through online forms and interviews with structured questions. This data helps fill dropdown menus and improves

the user experience. Shop information, such as names and services offered, is added manually by the admin via an input form and shown on the dashboard and shop locator for customers to see. Location data comes from Google Maps API and manual GPS input, enabling the system to show nearby partner shops on an interactive map. SMS notification logs are recorded automatically through the SMS gateway, keeping track of all service updates sent to customers. Finally, system feedback is gathered from users through surveys with Likert-scale questions, and the results are analyzed using simple statistics like averages and counts. This feedback helps measure satisfaction and system performance, guiding future improvements. Together, these data sources create a responsive, accurate, and user-focused platform for auto repair businesses and their customers.

CHAPTER 4

DESIGN AND METHODOLOGY

This chapter presents the design and methodology applied in developing the Auto Repair hub system. It includes the requirements analysis, documentation, system design with supporting diagrams, development methodology, and evaluation procedures.

Requirements Analysis

The Auto Repair Hub is a web-based system designed to streamline auto repair shop operations. Key features include user registration, online service booking, a real-time service queue tracker, and GIS mapping to locate nearby shops. Customers can book services, track their queue status, and receive updates. Shop owners manage bookings, update service progress, and handle customer interactions. System administrators oversee user management, update GIS data, monitor inventory, and generate reports. The system must be user-friendly, responsive across devices, secure, and scalable, with support for multiple users and role-based access control. These features aim to improve efficiency and provide a better experience for both customers and staff.

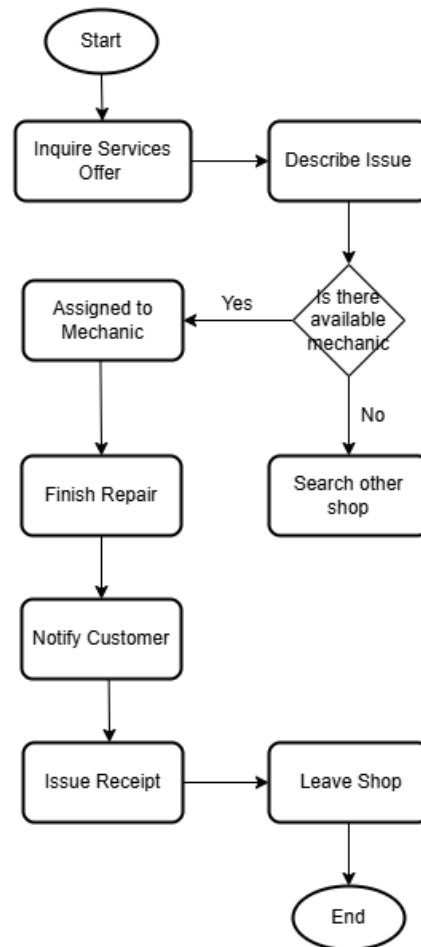


Figure 2. Existing Process Flow

Figure 2 Existing Process Flow Auto Repair Shops in Virac, Catanduanes typically operate using traditional, manual methods for managing their daily operations. The existing process often begins with walk-in customers inquiring about services directly at the shop. Customers explain their vehicle issues in person, after which the shop owner or available staff conducts a verbal assessment. There is usually no formal booking system,

which means customers may have to wait in line or return on another day if mechanics are fully occupied. This often leads to customer dissatisfaction due to uncertainty about waiting times and service availability. Assignment of work to mechanics is usually done manually by the shop owner or manager, based on availability and familiarity with the type of repair needed. There is minimal documentation, and job tracking is often done verbally or through handwritten notes. This lack of systematic record-keeping can result in inefficiencies such as delayed repairs, forgotten requests, or miscommunication between staff and customers. Once a repair is completed, customers are either contacted via personal phone calls or are expected to return at an estimated time to check on the progress. There is no structured notification system to alert customers in real time. Payment and receipt issuance are also handled manually, with receipts sometimes handwritten or, in some cases, not issued at all. These manual practices, while manageable in smaller setups, become limiting as the volume of customers increases. Furthermore, shop visibility and customer reach are limited. Many residents of Virac rely on word-of-mouth recommendations or physical signage to locate repair services. Without an online presence or a digital locator, potential customers, especially those unfamiliar with the area, may find it difficult to discover and compare shops based on location, service offerings, or reviews.

In this case, the current auto repair shop process in Virac lacks automation, proper scheduling, and efficient communication. This presents significant opportunities for improvement through the adoption of a web-based platform like the *Auto Repair Hub*,

which can modernize operations, improve customer experience, and streamline service delivery across local auto repair businesses.

Requirements Documentation

The system requirements were documented using various software engineering models such as use cases, user stories, and technical specifications. This documentation helps ensure all features are well-defined and understood by both developers and stakeholders.

Table 5. Functional Requirements

ID	Function	Description	Restriction
1	User registration/Login	Allow users (customer, staff, and admin) to create accounts and log in.	Type: Customer, Staff, Admin
2	Service Booking	Enables customers to select a shop, choose a service, and schedule appointments.	Type: Customer
3	Service Tracker Queue	Displays the status of ongoing services (Pending, In progress, Completed)	Type: Customer, Staff
4	Shop Locator via GIS	Provides a map interface showing nearby partner shops.	Type: Customer
5	Booking Management	Allows staff to view, confirm, update, or cancel customer bookings.	Type: Staff
6	SMS Notification System	Sends real-time SMS alerts for booking confirmations, status updates, and completions.	Type: System (automated)
7	Admin Dashboard	Provides admin with analytics, total bookings, registered users, and status overview	Type: Admin

ID	Function	Description	Restriction
8	User Management	Enables admins to manage (approve/deactivate) user and assign roles	Type: Admin

Table 5 describes the system's Functional Requirements, which include features like user registration and login, booking management, service tracker queue, GIS-based shop locator, SMS notifications, and role-based dashboards. These features aim to simplify the daily work of auto repair shops. Several factors may affect how these requirements are implemented and how well they work. For example, the system is initially limited to three partner shops and does not support online payments. Access to features depends on user roles: customers can only make one booking per time slot and cannot update service statuses, while staff members can manage bookings only for their assigned shop. The GIS shop locator uses manually entered location data, which may impact the accuracy of the maps. Users are expected to have basic digital skills to use the system smoothly, including filling out booking forms, tracking queues, and using interactive maps. Staff members need to regularly update service records and booking statuses to keep tracking accurate. The SMS alert system is designed to work automatically, assuming proper integration and user registration. Some parts of the system rely on external services and active user participation. The GIS locator uses Google Maps API to show shop locations, and SMS notifications are sent through a third-party service like Semaphore. The real-time service queue depends on continuous staff updates and stable backend operations to work correctly.

Table 6. Non-Functional Requirements

Criteria	Description
Functional Suitability	Provides essential features like booking, tracking, and service tracker queue.
Performance Efficiency	Fast response time and smooth handling for multiple users.
Compatibility	Works on major browsers and adapts to various screen sizes.
Interaction Capability	User-friendly interface for staff and customers.
Reliability	High availability with minimal downtime.
Security	Secures user data with authentication and encryption.
Maintainability	Easy to update and expand due to modular design.
Flexibility	Supports adding features and adjusting system settings.
Safety	Protects data with back up, recovery and access control.

Table 6 shows the Non-Functional Requirements, which cover key operational aspects like security, interface responsiveness, cross-browser compatibility, and system reliability. System responsiveness and availability can be influenced by the user's device capabilities and internet speed, especially in areas with unstable connections. Since the system is designed for modern browsers, users on older platforms may encounter compatibility issues. Offline access is not available, which could be a limitation for some users. It is expected that users will access the system using updated browsers and capable devices. Administrators are responsible for maintaining security by performing backups and monitoring system usage. The user interface supports both Filipino and English to

enhance accessibility for local users. Some system components depend on services provided by external vendors. GIS and SMS features rely on stable APIs and infrastructure from Google and Semaphore, respectively. Hosting servers must provide consistent uptime and quick data retrieval to meet performance standards. Future scalability will depend on the backend's ability to handle increased usage and additional partner shops. By taking these operational factors into account during development, the system can better adapt to real-world limitations, technical conditions, and user habits while maintaining its core performance goals.

Use Case Diagram

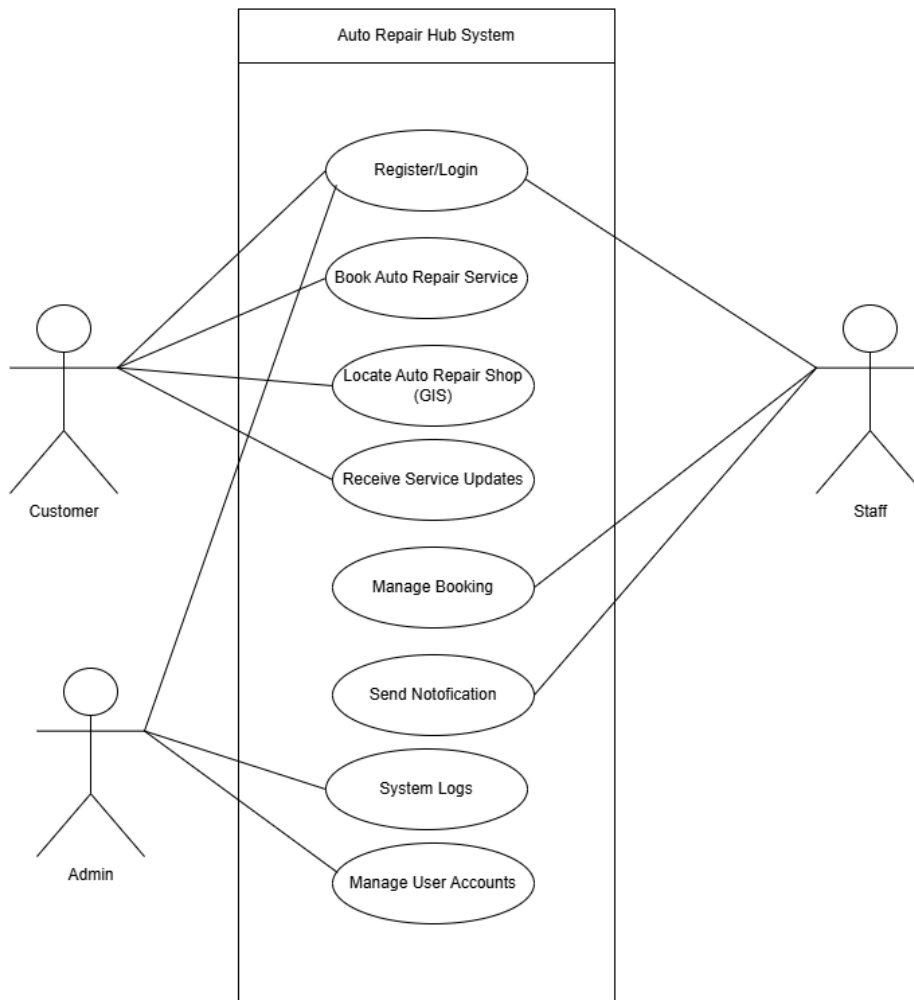


Figure 3. Use Case Diagram

Figure 3 The use case diagram shows the different actions users can take on the Auto Repair Hub platform. Customers can book appointments, track their vehicle repairs, and get notifications. Admins or shop owners can manage users, assign mechanics to jobs, update inventory, and oversee the workflow. Admin update service logs, check their

tasks, and provide repair status updates. These clear roles and actions help the system meet the needs of everyone involved, improving coordination, accountability, and service quality in the auto repair shop.

Design of Software, Systems, Product and/or Processes

This section includes visual models and diagrams that illustrate how the system is structured and functions. These are essential in guiding developers and communicating the system's functionality to stakeholders. The Auto Repair Hub is a web-based system that helps manage auto repair services more efficiently. It includes features like service booking, a real-time service tracker, a map to find nearby shops using GIS, and tools for managing inventory and reports. The system is built using three layers: the user interface (where users interact), the server (which handles the logic), and the database (where information is stored). There are three main user roles: customers, admins, and system administrator. Customers can sign up, book services, and track the status of their vehicles. Admins handle bookings, manage parts and users, and generate reports. Technicians update service progress in the system. The system starts with customer registration, followed by booking a service and selecting a shop from the map. The customer is placed in a queue, and the technician updates the service progress. Notifications are sent to the customer until the service is completed. This setup helps make auto repair services faster, clearer, and easier to manage.

Context Diagram

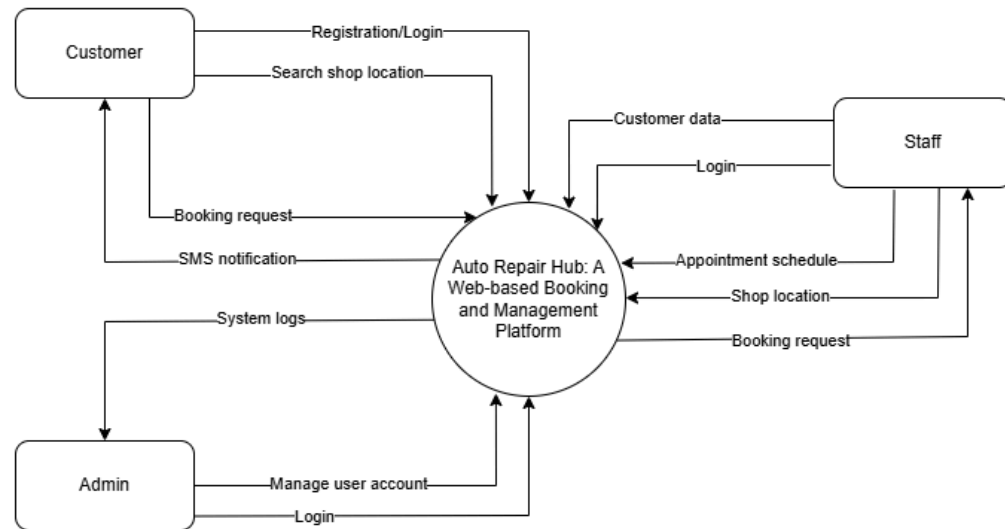


Figure 4. Context Diagram

Figure 4 shows the Context Diagram of the Auto Repair Hub, giving a clear overview of how the system interacts with outside parties. The main actors are Customers, Staff, Admin, and an SMS Gateway, with the diagram displaying how information flows between them and the central system. Customers are important users who register, log in, and make service bookings through the platform. After a booking is made, customers get updates via automated SMS messages sent by the system. These steps help keep customers informed and ensure their bookings are properly recorded and tracked. Shop staff use the system to approve or reject booking requests, update the status of services (like Pending, In Progress, or Completed), and keep an eye on the service queue. They also maintain current job records, which the system uses to send accurate notifications and create dashboard reports. Admins oversee the system by managing user

accounts, approving staff access, monitoring service activities, and producing performance reports. They can also adjust system settings, playing a key role in smooth operation and administration. The SMS Gateway is an external service linked to the system to send real-time notifications to customers. It helps keep users informed about their booking status, even in areas with poor internet access, which is especially important for the rural area of Virac, Catanduanes. This context diagram provides a clear picture of the Auto Repair Hub's boundaries and shows how each actor exchanges data with the system, supporting the goal of automating processes and digitally transforming local auto repair services.

Data Flow Diagram

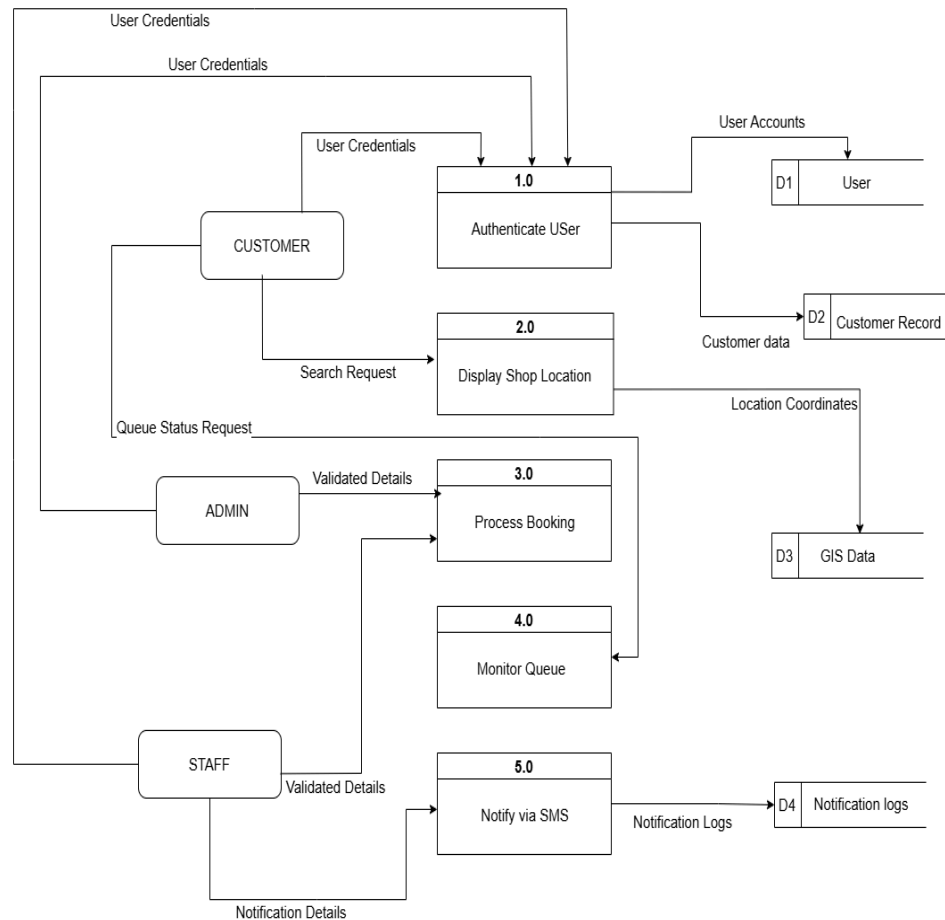


Figure 5. Data Flow Diagram Level 1

Figure 5 shows the Level 1 Data Flow Diagram (DFD) for the Auto Repair Hub, breaking down the system's internal processes into detailed functions. This DFD helps explain how user data is collected, processed, stored, and used throughout the platform. One key process is User Registration and Login, where customers, staff, or admins enter their credentials or create new accounts. The system checks this information, stores it

securely in the database, and grants access based on the user's role. Another important process is the Booking Management System. Here, customers enter their service preferences, pick a shop from the GIS-integrated map, and select a time slot. The system checks availability, creates a job order, and makes it visible to both staff and admins. The Service Queue Tracker monitors the status of all bookings. Staff update the queue in real time based on work progress. These updates flow through the system, keeping the queue status current and providing transparency for customers. The SMS Notification System works with both booking and queue processes. When an appointment is confirmed or a job's status changes (for example, to In Progress or Completed), the system automatically sends a request to the SMS Gateway so customers get timely alerts. The Admin process covers User Management and System Analytics. Administrators review user data, approve or deactivate accounts, and monitor performance using metrics like total bookings, active users, and service outcomes. All these processes connect to the central database, ensuring every action is logged, tracked, and easy to retrieve. This diagram offers a clear view of how the system works internally, making sure user interactions are handled smoothly and securely.

Process Swimlane

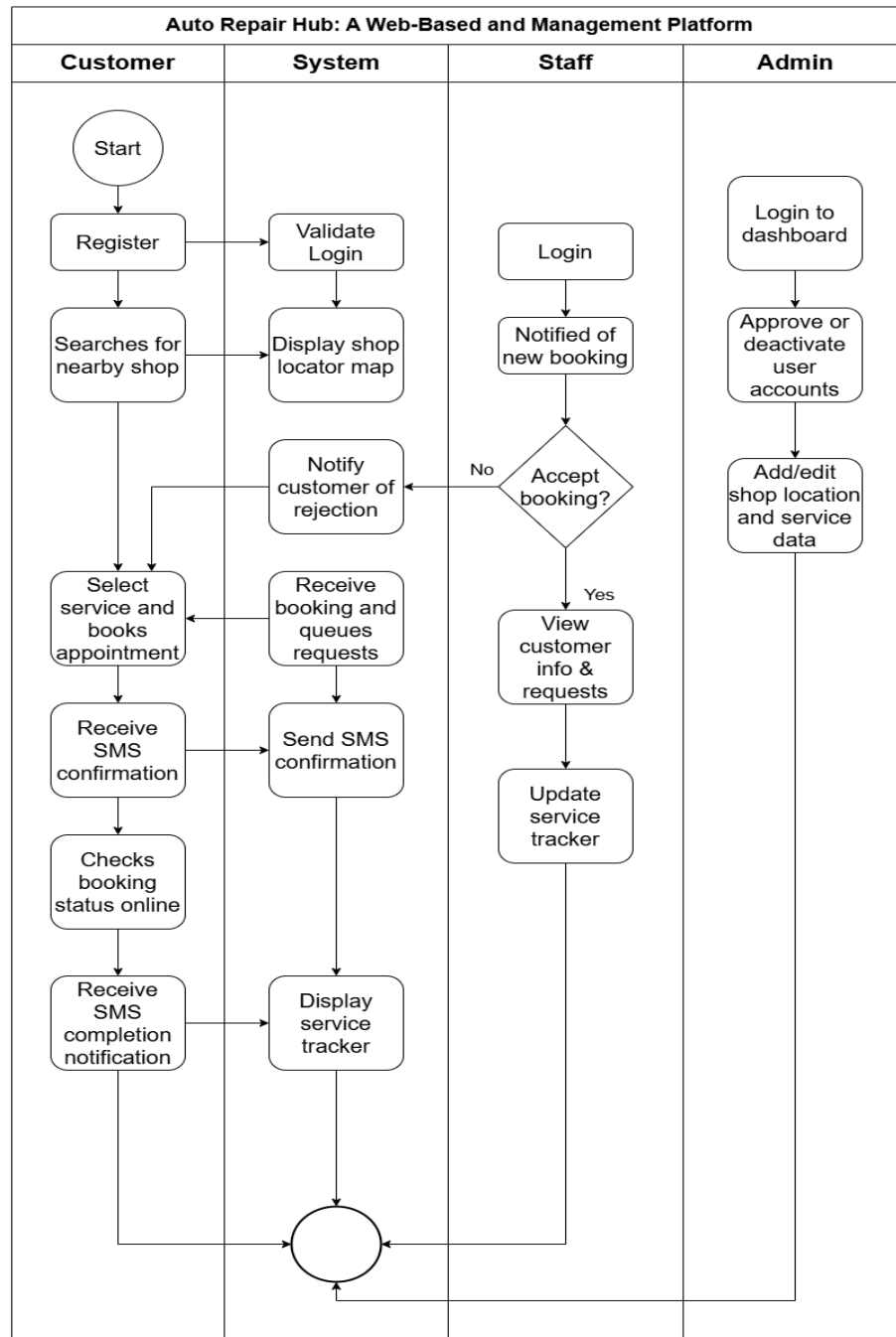


Figure 6. Process Swimlane (New Process)

Figure 6 shows the Process Swimlane Diagram for the Auto Repair Hub. This diagram visualizes the new digital workflow by categorizing system actions across three main actors: Customer, Staff, and System. It clarifies task responsibilities and the sequence of operations in a digitally transformed environment. The Customer lane begins with account registration or login. Once authenticated, the customer navigates the interface to locate nearby shops using the integrated GIS map and initiates a service booking. They input vehicle details, select the desired service, choose a preferred schedule, and confirm the booking. The Staff lane starts once a booking request is submitted. The system notifies the assigned shop personnel, who then review the request. They can accept or reschedule the booking based on availability. As the service progresses, staff update the status (Pending → In Progress → Completed). These updates ensure accurate tracking for both customers and administrative purposes. The System lane handles the automation of background processes. It manages booking validation, stores job order data, logs queue status changes, and sends SMS updates at each stage. For example, once a booking is confirmed or a job is completed, the system triggers real-time alerts to notify customers without requiring manual intervention. This swimlane diagram showcases the streamlined service workflow made possible by the Auto Repair Hub. It reduces redundancy, eliminates manual record-keeping, and enhances the customer experience by maintaining constant visibility and communication. By clearly delineating responsibilities and system roles, it improves operational efficiency, accountability, and service quality across the shop's digital processes.

Entity-Relationship Diagram / Database Schema

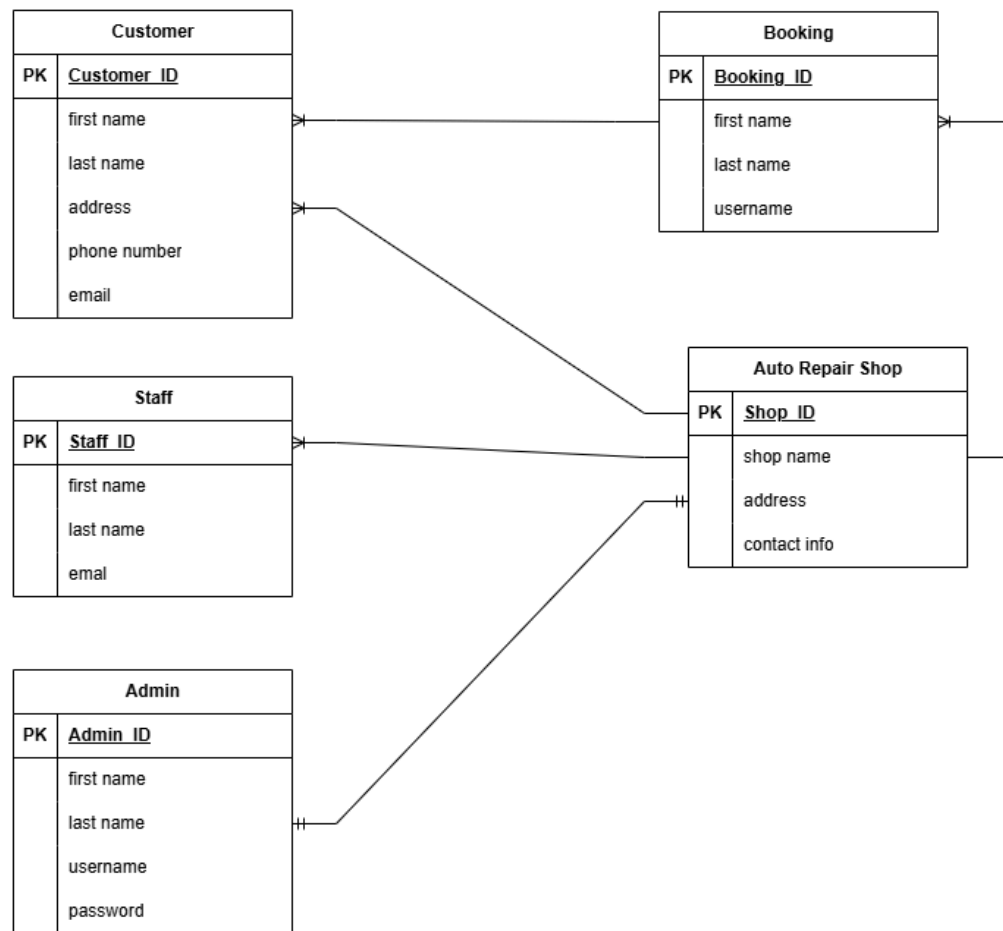


Figure 7. Entity-Relationship Diagram / Database Schema

Figure 7 Entity Relationship Diagram for Auto Repair Hub shows how the system's data is organized and connected. It includes main entities like Users, Bookings, Services, Inventory, and Shops, all linked to keep data flowing smoothly between different parts of the system. Customers are part of the Users group, each with their own

details and roles. Bookings connect customers to their scheduled services, while Service Logs keep track of completed jobs and mechanic activities. This well-structured database helps with accurate data access, efficient reporting, and growth, which are all important for running a successful auto repair business.

System Architecture

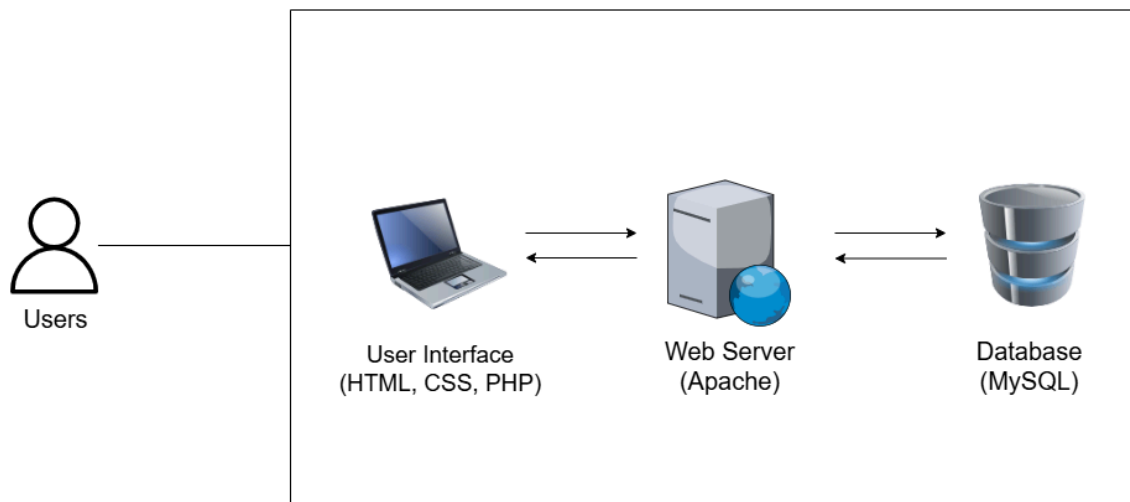


Figure 8. System Architecture

Figure 8 shows the system architecture of the Auto Repair Hub, which uses a strong three-tier model made up of the Presentation Layer, Application Layer, and Data Layer. This setup helps the system grow, stay easy to maintain, and offer a smooth user experience. The Presentation Layer is the user interface of the system. It is built with HTML, CSS, JavaScript, and Bootstrap to make sure the design works well and is

accessible on desktops, tablets, and smartphones. This layer includes all visible parts like registration forms, booking screens, live service dashboards, and interactive maps. It ensures users can easily use the platform, no matter their technical skills. The Application Layer is the system's main engine. Created with PHP, it manages all backend processes and workflows. This layer handles user inputs, checks booking schedules, manages login sessions, calculates queue positions, and sends SMS notifications using the Semaphore API. It also connects with the Google Maps API to show real-time locations and help select shops. The Application Layer links the frontend interface with the backend data, making sure business rules are followed correctly. The Data Layer uses MySQL as the database system. It stores all important data like user credentials, service bookings, shop information, queue status, and SMS records. It is designed with clear relationships to allow fast data access and keep data accurate. Access controls are in place to limit sensitive data based on user roles. This layered architecture supports modular development, letting each part be updated or scaled on its own. For example, future features like mobile apps, payment gateways, or analytics dashboards can be added without interrupting the current system. By separating tasks, the design improves system speed and security while making maintenance and debugging easier.

GIS Architecture

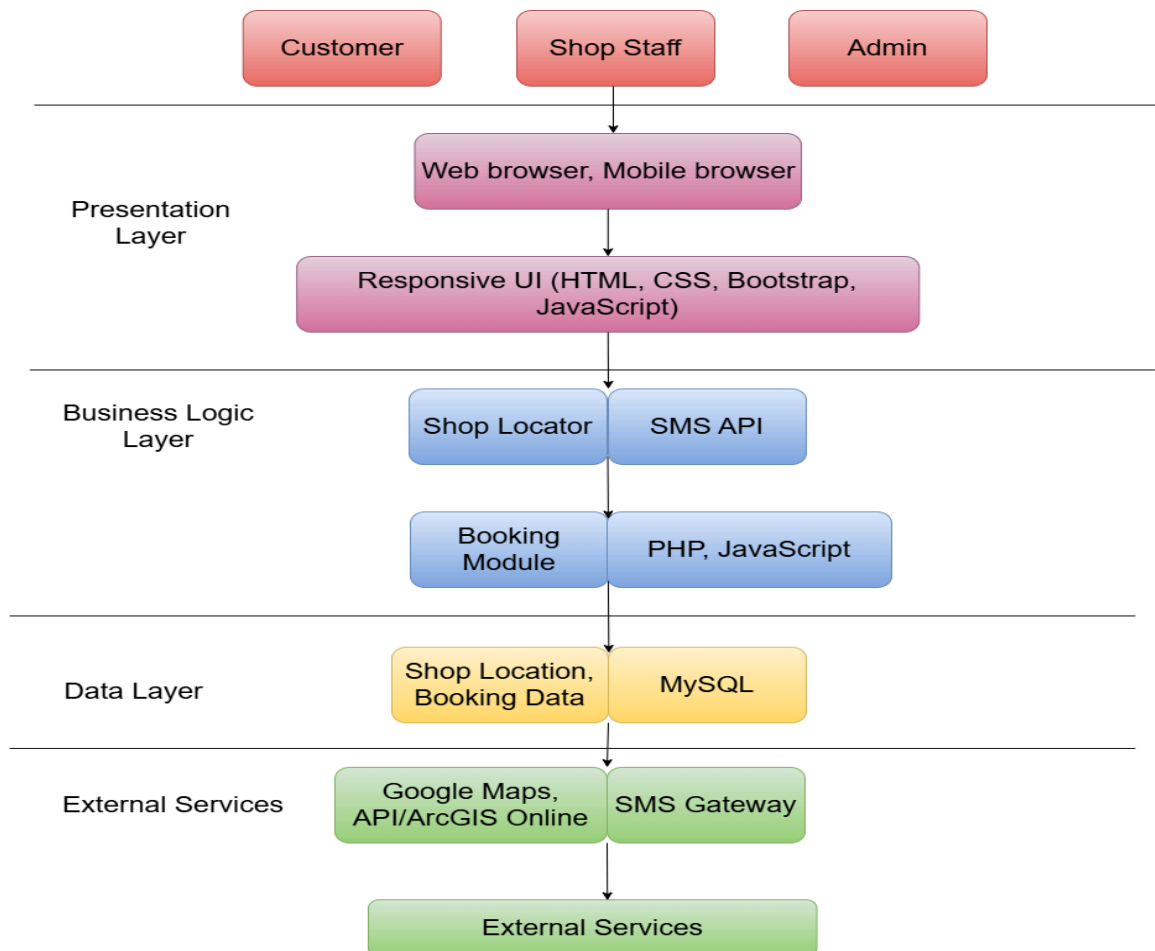


Figure 9. GIS Architecture

Figure 9 shows the GIS Architecture as part of the Auto Repair Hub. This framework is built to manage spatial data and offer interactive location-based services. At the base are the GIS data sources, which include manually entered coordinates and real-time data from the Google Maps API. This information is stored and managed in the

system's MySQL database. The heart of the GIS system is the Mapping Engine, which uses tools like the Google Maps JavaScript API or Leaflet.js to turn raw coordinate data into interactive maps. On the user interface, customers see a live map with pins marking partner shops. Clicking a pin shows the shop's name, services, address, and a link to the booking form. Meanwhile, the admin tools let backend users manage GIS data by adding new locations, updating coordinates, or removing outdated entries. This setup offers dynamic maps and makes the service easier to access, especially for customers unfamiliar with Virac's geography. The GIS layer not only makes navigation simpler but also helps users find services and make better decisions.

Storyboard

The storyboard provides a visual guide showing how different users, customers, staff, and administrators interact with the Auto Repair Hub system. It lays out the steps users take and how the system responds, offering a clear and easy-to-understand look at how the platform works. Each screen in the storyboard highlights key processes like finding a repair shop using GIS, booking services, tracking service status, managing appointments, and receiving updates. By walking through these user journeys step by step, the storyboard connects the design ideas to real-world use. It also helps everyone involved developers, evaluators, and users see how the system functions in daily situations, ensuring it is user-friendly, clear, and meets the project's goals.

Customer Side



The image shows a user interface for a login and registration system. It consists of a large outer rectangle containing a smaller inner rectangle. Inside the inner rectangle, the text "Have an account?" is centered at the top. Below this text are three input fields, each preceded by a label: "Register:", "Login:", and "Password:". These labels and their corresponding input fields are aligned to the left. At the bottom center of the inner rectangle is a rounded rectangular button labeled "Confirm".

Figure 10. Customer Login

Figure 10 Customer Login before booking, users must log in or register on the Login/Registration Page. New customers can create an account, while returning users log in with their username and password. This page ensures secure access to personal information and service history. The confirm button completes the login, linking bookings to the user's account.



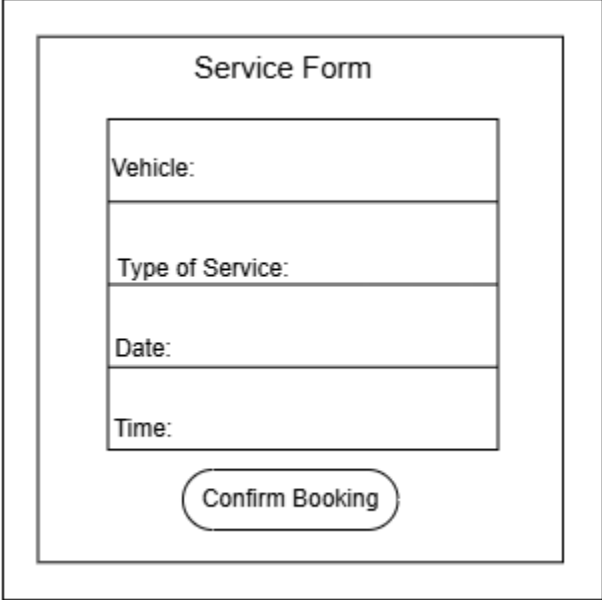
Figure 11. Homepage

Figure 11 Homepage serves as the main landing page for visitors to Auto Repair Hub. It has a simple and friendly design focused on helping customers “Find a Repair Shop Near You.” There is a single “Find” button that starts the process, using GPS to show nearby auto repair shops. This straightforward layout encourages users to engage quickly and access key services without hassle.



Figure 12. Shop Locator

Figure 12 Shop Locator when customers click the “Find” button, they are taken to the Shop Locator page. This page features a map showing nearby shops like Molje Lube, CT Gear, and Precision Tech Moto. Each shop is marked with a pin on the map, and clicking a pin displays the shop’s name, address, services offered, and a “Book a Service” button. This interactive map lets customers easily explore options based on location and available services, making it simple to choose the right shop.



The diagram shows a 'Service Form' interface. It consists of a large outer rectangle containing a smaller inner rectangle. The inner rectangle is titled 'Service Form' at the top. Inside this inner rectangle, there are four stacked input fields, each with a label to its left: 'Vehicle:', 'Type of Service:', 'Date:', and 'Time:'. Below these fields, centered within the inner rectangle, is a rounded rectangular button labeled 'Confirm Booking'.

Figure 13. Service Form

Figure 13 Service Form After logging in, customers reach the Service Form, where they enter vehicle details, select the type of service (like oil change or repaint), and choose a preferred date and time for the appointment. Once all fields are completed, clicking “Confirm Booking” submits the request. This form gathers important information to schedule and manage appointments accurately.

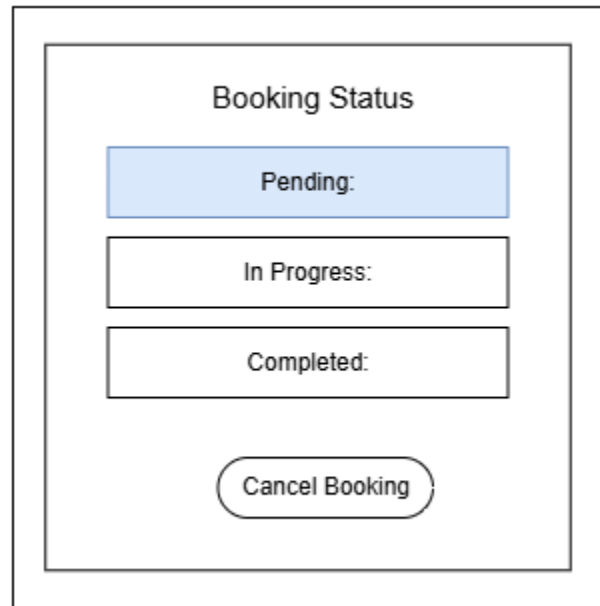
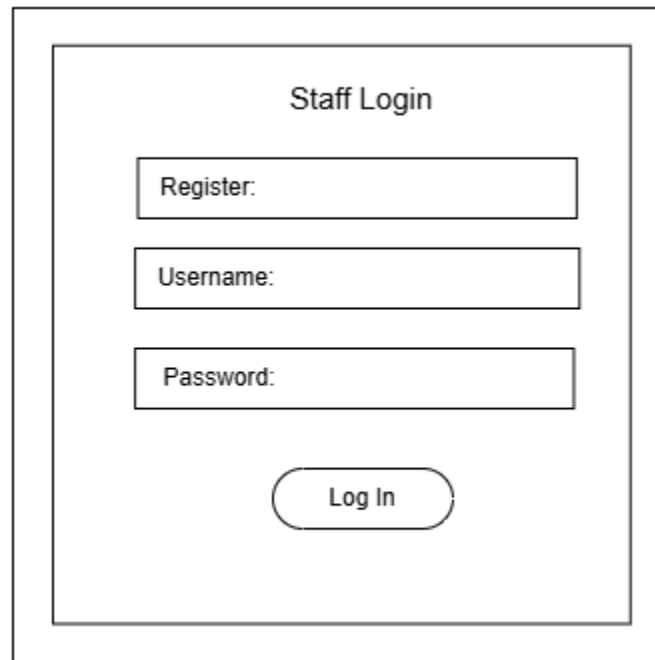


Figure 14. Booking Status

Figure 14 Booking Status Once a booking is confirmed, customers can check its progress on the Booking Status page. The status updates through three stages: Pending, In Progress, and Completed. Each booking changes status in real time as the service progresses. There is also a “Cancel Booking” button, allowing customers to cancel or reschedule if needed. This real-time tracking keeps customers informed and improves their experience.

Staff Side



The diagram illustrates a 'Staff Login' form. It is enclosed in a rectangular border. At the top center, the text 'Staff Login' is displayed. Below this, there are three input fields, each preceded by a label: 'Register:', 'Username:', and 'Password:'. These labels and their corresponding input boxes are aligned to the left. At the bottom center of the form, there is a rounded rectangular button labeled 'Log In'.

Figure 15. Staff Login

Figure 15 Staff Login offers a secure way for repair shop staff to access the system. Staff members can register or log in with their credentials. This separate login ensures only authorized personnel can use the tools to manage bookings and customer interactions. It supports secure access based on staff roles.

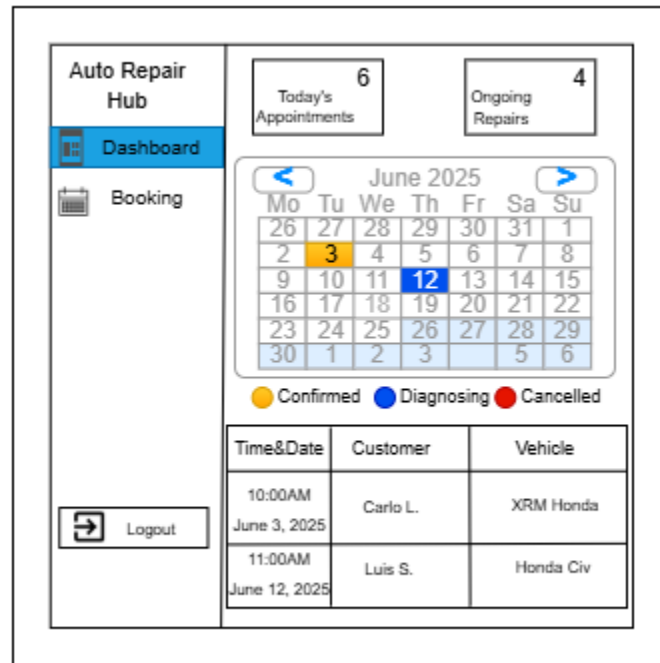


Figure 16. Admin Dashboard

Figure 16 Admin Dashboard after logging in, staff see the Admin Dashboard, a central page showing a calendar of appointments and status summaries. It includes counts of today's appointments and ongoing repairs. The calendar uses colors to show confirmed, ongoing, or canceled bookings. Below the calendar, each booking lists the time, date, customer name, and vehicle details. This dashboard helps staff organize their day, manage tasks, and prioritize work efficiently.

The screenshot displays the 'Auto Repair Hub' interface. On the left is a sidebar with a 'Dashboard' link and a 'Booking' link, the latter of which is highlighted in blue. Below the sidebar is a 'Logout' button. The main content area is divided into three sections. At the top, there are three status boxes: '1 Repair in Progress', '2 Pending', and '3 Completed'. Below these is a table titled 'Today's Appointments' with columns for 'Time&Date', 'Customer', and 'Status'. The table contains two rows: one for '10:00AM June 3, 2025' by 'Carlo L.' with a 'Completed' status, and another for '11:00AM June 12, 2025' by 'Luis S.' with a 'Diagnosing' status. At the bottom of the main area is a 'Booking Request' form for 'Carlo L.' with the service 'Oil Change' scheduled for 'June 3, 2025 at 10:00AM'. This form includes 'Cancel' and 'Confirm' buttons.

Today's Appointments		
Time&Date	Customer	Status
10:00AM June 3, 2025	Carlo L.	Completed
11:00AM June 12, 2025	Luis S.	Diagnosing

Booking Request

Carlo L.
Service
Oil Change

Date and Time
June 3, 2025 at 10:00AM

Figure 17. Booking Request

Figure 17 Booking Request in this section, staff can review and handle new service requests. Each booking shows the customer's name, service type, scheduled time, and current status. Staff can update the status to "Pending," "In Progress," or "Completed," and confirm or cancel appointments as needed. This keeps the customer's booking status in sync with the repair workflow.

Admin Side



The diagram shows a rectangular box representing the Admin Login interface. Inside this box, there is a smaller rectangular area containing the text "Admin Login" at the top. Below the title, there are two input fields: the first is labeled "Username:" and the second is labeled "Password:". Below these fields is a rounded rectangular button labeled "Log In".

Figure 18. Admin Login

Figure 18 Admin Login this is for higher-level users who oversee the entire system. It has secure username and password fields. Admins have more privileges than regular staff, including managing users, viewing analytics, and configuring the platform.

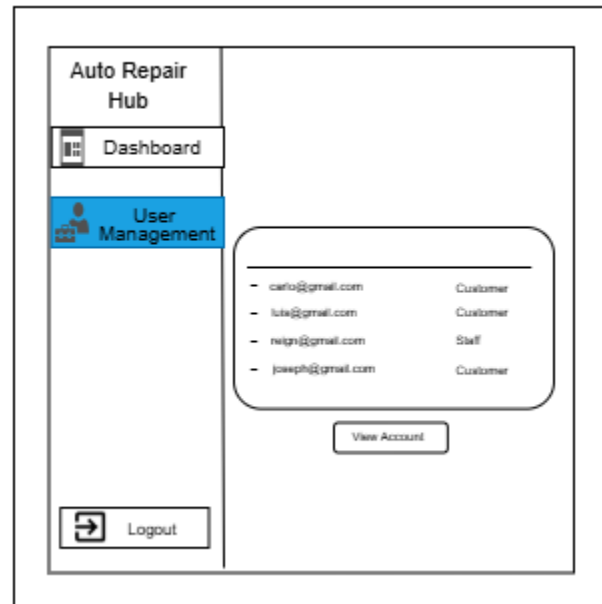


Figure 19. User Management

Figure 19 User Management one key feature for admins is the User Management Page. It lists all registered users both customers and staff, with their roles and account details. Admins can manage accounts, reset passwords, or view profiles. This helps maintain account security and handle any user issues.

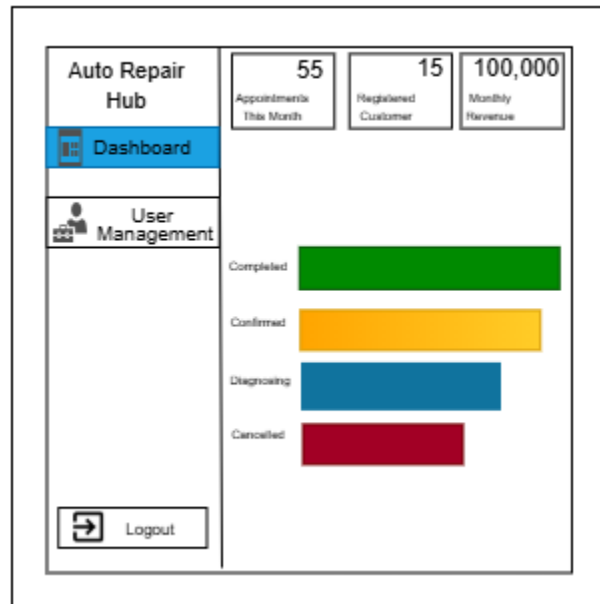


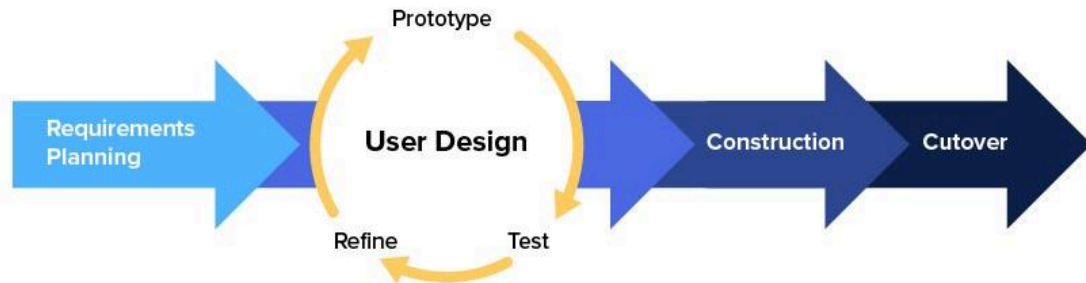
Figure 20. Admin Dashboard

Figure 20 Admin Dashboard gives visual reports on system usage. It shows important data like total registered customers, staff members, number of appointments, and booking status categories. The information is displayed through bar graphs and widgets for easy understanding. This helps admins make informed decisions and monitor performance trends.

Development and Testing

Software Development Method

The creation of the Auto Repair Hub was conducted using the Rapid Application Development (RAD) pattern, which supported rapid, user-focused development. It started from the requirements planning stage, when the team engaged with local automobile repair shop business owners, including Molje Lube, and discussed typical issues like manual scheduling and inadequate stock tracking, which informed the feature design of the system. This was then followed by the user design phase, wherein sample layouts and wireframes were developed and iteratively improved upon based on input from prospective users to guarantee usability and utility. In the construction phase, the system was actually implemented, incorporating fundamental functions such as service scheduling, inventory management, and real-time alerts through suitable web technologies. Lastly, the cutover process included system rollout, testing via live user engagement, and carrying out training sessions to familiarize shop staff and customers with the platform prior to complete deployment.



ImageSource: <https://kissflow.com/hs-fs/hubfs/Rapid-application-development.png?width=850&height=450&name=Rapid-application-development.png>

Figure 21. Rapid Application Development

The RAD process in this study includes four main phases: Planning, User Design, Construction, and Cutover.

Planning Phase: The planning phase of the Auto Repair Hub project involves defining the project's objectives and determining its feasibility in terms of technology and budget. Key stakeholders, such as auto repair shop owners and mechanics, are identified to ensure the system meets their needs. Requirements are gathered through consultations to outline essential features like appointment scheduling, repair tracking, and invoicing, along with important considerations for security and usability. The project team, necessary tools, and resources are then planned, and potential risks are assessed with strategies developed to manage them. A detailed project timeline with milestones is

created and presented to stakeholders for approval, ensuring that the project proceeds in an organized and efficient manner.

User Design Phase: The User Design Phase of the Auto Repair Hub project focuses on close collaboration with end users, particularly shop owners and customers, to gather system requirements and define how the platform should function. During this phase, mockups of key features such as service booking, GIS-based shop locating, and the service queue tracker are developed. Users provide feedback on interface layouts and usability, allowing for quick refinements before development begins.

Construction Phase: Construction Phase, developers begin building the system based on the approved designs. This includes implementing core modules like user login, online booking, queue tracking, GIS mapping, and the admin dashboard. Throughout development, users participate in testing prototypes to identify bugs or usability issues early. Rapid iteration allows the system to evolve quickly and align closely with user expectations.

Cut-Over Phase: Cut-Over Phase marks the transition from development to live deployment. The system is installed in its operational environment, and any existing manual data is inputted. Shop owners and system administrators receive training to use the platform effectively. Minor adjustments are made as the system goes live to ensure smooth adoption.

Evaluation

The evaluation of the Auto Repair Hub used a descriptive approach with structured surveys based on ISO/IEC 25010 quality standards. These standards provide a worldwide benchmark for judging software quality in areas like functionality, usability, performance, reliability, maintenance, compatibility, security, flexibility, and safety. Participants rated each quality on a five-point scale from strongly disagree (1) to strongly agree (5). The responses were analyzed using frequency counts, percentage distribution, and averages to summarize user satisfaction. This approach allowed the research team to measure how well the platform met user needs and to pinpoint areas that could be improved. In addition, open-ended survey questions and follow-up interviews offered more detailed feedback about the system's strengths and weaknesses. Together, these methods gave a complete and user-based evaluation of the platform's effectiveness.

Data Gathering Procedure

To thoroughly evaluate the Auto Repair Hub system, a structured data collection process was put in place, focusing on system usability, performance, and user satisfaction. It began with consultations and observations at selected partner repair shops to understand their workflows and identify challenges related to service scheduling, job tracking, and customer communication. These findings directly shaped the system's design and development priorities. During testing, prototype versions were given to

representative users, including shop owners, staff, and customers. They were asked to complete key tasks such as booking services, tracking service status, and using the GIS feature to locate shops. Their experiences were assessed using tools based on the ISO/IEC 25010 software quality model. Surveys with Likert scale questions measured user opinions on functional suitability, usability, performance efficiency, and maintainability. The responses were analyzed using descriptive statistics, like frequency counts and average scores, to evaluate overall acceptance and highlight areas for improvement. Additional feedback came from informal interviews and user comments during system use, providing deeper insights into user interaction and satisfaction. This evaluation-focused approach ensured that the Auto Repair Hub was developed around real user needs and assessed according to recognized international software quality standards.

Respondents of the Study

The study used ***purposive sampling*** to select individuals who were the primary users and evaluators of the Auto Repair Hub system. This approach ensured that participants had relevant experience and knowledge to provide valuable feedback on the system's usability, functionality, and technical quality. Three shop owners from partner repair shops Molje Lube, Precision Tech Motoshop, and CT Gear were included to share managerial insights on shop operations, service coordination, and booking management.

10 customers, chosen for their regular use of these shops' services, evaluated the platform's usability, convenience, and overall satisfaction. Additionally, three IT professionals were selected as technical experts to review the system's architecture, interface design, security features, and integration quality. Since these experts were chosen based on their qualifications rather than a larger population, no population estimate was given for this group. This multi-perspective approach provided a comprehensive evaluation of the system, combining operational, user, and technical viewpoints.

Table 7. Respondents of the Study

Type of Respondent	Total Population	Sample
Shops Owners		3
Customer		10
Technical Experts	3	3
Total		16

Table 7 shows the breakdown of respondents chosen through purposive sampling to evaluate the Auto Repair Hub system. The sample included three main groups: shop owners, customers, and technical experts. Among many shop owners in Virac, Catanduanes, three were selected as direct partners Molje Lube, Precision Tech Motoshop, and CT Gear. These participants provided valuable insights into how the system fits with real shop operations, such as job order management, service tracking, and customer coordination. Ten customers who regularly use services from these partner shops were selected based on their experience and familiarity with both traditional and

new service processes. Their feedback focused on the system's ease of use, convenience, and overall satisfaction with the digital platform. Additionally, three technical experts in information systems and software development were consulted to assess the system's design, security features, and compliance with ISO/IEC 25010 standards. These experts were chosen for their professional knowledge, so statistical generalization was not necessary. This combination of respondents administrators, end-users, and technical evaluators provides a well-rounded review of the Auto Repair Hub system. It allowed for a thorough evaluation of both practical usability and technical performance, helping the research team identify strengths and areas for improvement before full implementation.

Statistical Tool

To make the analysis clear and transparent, the following statistical formulas were used. For frequency count, we tallied the number of respondents who chose each response option to see how often certain perceptions or behaviors appeared. This approach is especially helpful for summarizing categorical data and spotting main trends.

For percentage calculation, the formula:

$$\text{Percentage} = \left(\frac{\text{Frequency}}{\text{Total Number of Respondents}} \right) \times 100$$

It was used to show how much each response makes up the whole sample. This makes it easier to compare different response categories, especially when working with scaled feedback.

To calculate the mean, or average score, of responses to Likert-scale questions, the formula is:

$$\text{Mean: } \frac{\sum X}{N}$$

The calculation was done using the formula where $\sum X$ represents the sum of all respondents' scores, and N is the total number of respondents. The mean helps to find the central tendency of opinions or satisfaction levels, providing insight into the overall perception of system usability or performance.

The Likert scale used usually ranges from 1 (Strongly Disagree) to 5 (Strongly Agree). Each response is given a numerical value, and the scores are interpreted according to a set scale. For example:

Table 8. Likert Scale

Range	Quantitative Equivalent	Qualitative Equivalent
4.21 - 5.00	5	Very Satisfied
3.41 - 4.20	4	Satisfied

2.61 - 3.40	3	Neutral
1.81 - 2.60	2	Dissatisfied
1.00 - 1.80	1	Very Dissatisfied

This interpretation scheme offers a clear and measurable way to assess respondents' satisfaction, the system's effectiveness, and its usability. Together, these statistical tools provide a strong evaluation of the Auto Repair Hub, ensuring the analysis aligns with the study's goals.

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